Independent Mathematical Review

FIBER-GRAND Paper I: Exact One-Deletion Decoding

3 August 2026

Overall mathematical verdict: PASS_WITH_EDITS
Correctness effect: No identified defect changes any theorem statement, the exact-ML guarantee, the complete tie-set guarantee, the finite-length query bounds, or the high-probability exponent.
Required edits: Two proof-writing repairs are required: (i) split off finite exhaustion before invoking the unseen-candidate bound at shell $s = T + L_n - 1$; and (ii) replace an unjustified asymptotic equality in the supplement by a monotonicity-and-continuity upper bound. A terminology clarification concerning "history" is also advisable.
Scope effect: None. The conclusions remain restricted to one uniformly located deletion, independent hard BSC substitutions, a uniform prior on an arbitrary nonempty binary membership-testable code, and membership/query accounting.

1 Materials, integrity, and review basis

The reviewed archive was

`FIBER_GRAND_Paper_I_External_Review_Bundle_Phase2D_R1.zip`.

Its SHA-256 value was independently recomputed as

`a485bbf4f655f73b4250f9048a7b913b07db8d25000f6d8737f06373c0d02cd0`,

which matches the supplied value. Every entry listed in `BUNDLE_MANIFEST.sha256` also verified.

The mathematical review used primarily:

1. `01_Conference_Manuscript.tex` and its four-page rendered PDF;
2. `02_Proof_Supplement.tex` and its four-page rendered PDF;
3. `04_EXTERNAL_PROOF_REVIEW_CHECKLIST.md`; and
4. `FIBER_GRAND_Research_Proposal_rev4_scientific_audit.tex`.

The proofs were reconstructed directly. Independent finite enumerations and exact rational checks were used only as corroborating sanity checks, not as substitutes for proof. This report is a mathematical-correctness review; it does not certify novelty, patentability, or venue significance.

For compact source references below, **M** means the conference manuscript, **S** the proof supplement, and **P** the research proposal. Line numbers refer to the corresponding source files in the supplied materials.

2 Scientific review of the broader proposal

2.1 Scientifically sound core

The proposal's central premise is correct and important: on a non-injective channel, maximum-likelihood decoding ranks candidate inputs by

$$W(y \mid x) = \sum_{z:T_z(x)=y} p(z \mid x),$$

not by the probability of only the largest latent path. Paper I supplies a fully explicit realization of this premise. In the one-deletion+BSC model, the candidate likelihood is a sum over deletion alignments, and the strict family in Theorem 1 demonstrates that the largest single alignment component and the aggregate candidate likelihood can order two codewords oppositely.

The proposal is also scientifically disciplined in separating code-modular correctness from practical efficiency. Membership-only access is sufficient for exactness, but the cost of membership is code-dependent and must not be treated as universally constant. The proposal explicitly treats the weighted-transducer, bounded-edit, tractability, soft-output, and fiber-guesswork directions as targets rather than established results.

2.2 What Paper I establishes, and what it does not

Paper I establishes the first narrow milestone of the proposal:

- an exact likelihood for one uniform deletion plus independent substitutions;
- a strict path-component/aggregate-ML reversal family;
- an exact inverse-alignment shell decoder for every nonempty binary membership-testable code;
- complete ML tie-set certification from a valid unseen-candidate upper bound;
- a code-independent realization-dependent stopping-shell bound;
- deterministic, finite-confidence, expected, and high-probability membership/query upper bounds.

It does *not* establish the proposal's desired general structural characterization of all first-hit-compatible channels, tightness or optimality of the exponent, a lower bound for all exact code-agnostic decoders, multiple-edit tractability, transducer-scale practicality, or telecommunications-system advantage. Those remain legitimate research objectives. The present $h_2(p_s)$ upper exponent is useful but follows from a direct Hamming-volume argument; by itself it is not the proposal's long-term "field-defining" theorem.

2.3 Proposal revisions indicated by the completed Paper I

The proposal is scientifically promising and worth continuing, but it should be synchronized with the stronger Paper-I formulation.

1. **Move membership before complete scoring in the one-deletion baseline.** P lines 638–645 compute the complete likelihood before calling `IsCodeword`. Paper I correctly queries membership first and completely scores only discovered codewords. The proposal should adopt that order for the uniform-deletion/BSC baseline.
2. **Replace the old cost vocabulary by the Paper-I separation.** P lines 654–661 use Q_{cand} in a way that can mix discovered ambient candidates and fully scored codewords. The achieved result requires distinct counters for generated hypotheses, globally deduplicated membership calls, complete codeword-likelihood evaluations, and wall-clock cost. The relevant comparison is $|\mathcal{C}|/Q_{\mathrm{score}}$, not a universal latency ratio.

3. **Present the shell decoder as the completed baseline and retain the multi-frontier construction as a generalization.** The shell certificate is sharper and easier to audit for the uniform one-deletion+BSC model. The proposal's sum-of-frontiers certificate remains valid as a broader architecture for nonuniform alignments or heterogeneous component streams.
4. **Retain the staged impact claim.** The proposal is justified as a research program, but broad impact still depends on a structural boundary theorem, a meaningful tractability or lower-bound result, and a calibrated system regime. Paper I alone supports a narrow exact-decoding contribution, not the entire long-term claim.

Proposal scientific verdict: continue. The core mechanism is valid and Paper I supplies genuine positive evidence. The broader program remains high-risk but properly falsifiable.

3 Independent reconstruction of the main proof chain

3.1 Channel likelihood and sliding recurrence

Let $m = n-1$ and let substitutions occur before deletion. Conditioned on deleting coordinate j, the m surviving substitution variables are fixed by (x, y, j) and contribute

$$p_s^{d_j(x,y)}(1-p_s)^{m-d_j(x,y)}.$$

The substitution variable at the deleted coordinate is unobserved, and summing its two values produces

$$(1-p_s)+p_s = 1.$$

Averaging over a uniform deletion coordinate therefore gives

$$W(y \mid x) = \frac{1}{n}\sum_{j=1}^{n} p_s^{d_j(x,y)}(1-p_s)^{m-d_j(x,y)}.$$

Moving the hypothesized deletion from j to $j+1$ removes the old comparison (x_{j+1}, y_j) and adds the new comparison (x_j, y_j), hence

$$d_{j+1} = d_j - \mathbf{1}\{x_{j+1} \neq y_j\} + \mathbf{1}\{x_j \neq y_j\}.$$

Both formulas are correct under the manuscript's coordinate convention.

3.2 Strict best-component/aggregate-ML reversal

Write $q = 1-p_s$. For

$$y = 0^{n-3}10, \qquad x_A = 0^{n-3}101, \qquad x_B = 0^n,$$

the alignment distances of x_A are

$$0, 1, 2, \underbrace{3, \ldots, 3}_{n-3 \text{ times}},$$

whereas every alignment of x_B has distance one. For $n = 4$, the x_A distances are exactly $(3, 2, 1, 0)$, so the endpoint case is included.

The largest alignment components are

$$M_y(x_A) = \frac{1}{n}q^{n-1}, \qquad M_y(x_B) = \frac{1}{n}p_s q^{n-2},$$

and $M_y(x_A) > M_y(x_B)$ because $q > p_s$. The aggregate likelihoods are

$$W(y \mid x_B) = p_s q^{n-2},$$
$$W(y \mid x_A) = \frac{1}{n}\left[q^{n-1} + p_s q^{n-2} + p_s^2 q^{n-3} + (n-3)p_s^3 q^{n-4}\right].$$

Direct expansion gives

$$\frac{n\{W(y \mid x_B) - W(y \mid x_A)\}}{q^{n-4}} = (1-2p_s)(np_s - 1 - 2p_s^2).$$

Thus the aggregate ordering reverses precisely when $np_s > 1 + 2p_s^2$. At equality the two aggregate likelihoods tie, while the best-component inequality remains strict. The smaller equality root is

$$p_s^\star = \frac{n - \sqrt{n^2-8}}{4},$$

which lies in $(0, 1/2)$ for every $n \geq 4$.

3.3 Discovery shells and unseen-candidate certificate

For a length-m binary word z, inserting bit b in the first gap and immediately after each symbol unequal to b generates every distinct one-insertion supersequence exactly once within the b class. Its size is

$$1 + \#\{i : z_i \neq 0\} + 1 + \#\{i : z_i \neq 1\} = m + 2 = n + 1.$$

The $b = 0$ and $b = 1$ classes are disjoint because their Hamming weights differ by one.

A candidate x is generated from shell s exactly when there is an alignment j such that deleting x_j yields $z = y \oplus e$ with $\mathrm{wt}(e) = d_j(x, y) = s$. Therefore its first shell is

$$h_y(x) = \min_j d_j(x, y).$$

After completing shell $s < m$, an unseen word has $d_j \geq s + 1$ for every alignment. Since $p_s/(1-p_s) < 1$, every alignment component is at most

$$U_s = p_s^{s+1}(1-p_s)^{m-s-1},$$

and so is their uniform average. If the incumbent score is strictly larger than U_s, no unseen codeword can beat or tie it. The strict comparison is therefore exactly what certifies the *complete* ML tie set.

A concrete interior-probability witness shows why replacing $>$ by $\geq$ is invalid for tie completeness. Let

$$n = 4, \quad y = 001, \quad p_s = \frac{1}{3}, \quad s = 0,$$

and take the code $\mathcal{C} = \{0010, 0000\}$. The word 0010 is seen in shell zero and has distance vector $(2, 2, 1, 0)$; the word 0000 is unseen and has distance vector $(1, 1, 1, 1)$. Both have likelihood

$$W(y \mid 0010) = W(y \mid 0000) = U_0 = \frac{4}{27}.$$

A non-strict stop after shell zero would miss an unseen ML tie.

3.4 Stopping shell and finite work

Let T be the realized number of substitutions among the m surviving symbols and let $\rho = p_s/(1-p_s)$. The exact integer

$$L_n(p_s) = \min\{\ell \ge 1 : \rho^\ell < 1/n\} = \left\lfloor \log_{(1-p_s)/p_s} n \right\rfloor + 1$$

is correct, including when n is an exact power of the logarithm base: the defining inequality is strict, so one additional integer step is required.

Under the true deletion alignment, the transmitted codeword is discovered by shell T and has aggregate score at least the single-component mass

$$\frac{1}{n} p_s^T (1-p_s)^{m-T}.$$

For $s_0 = T + L_n - 1 < m$,

$$\frac{U_{s_0}}{n^{-1} p_s^T (1-p_s)^{m-T}} = n\rho^{L_n} < 1.$$

Hence the strict certificate fires by s_0. If $s_0 \ge m$, finite exhaustion gives the claimed bound. Therefore

$$S \le \min\{m, T + L_n - 1\}.$$

Completing shells through S processes $\mathrm{Vol}(m,S) = \sum_{w=0}^{S} \binom{m}{w}$ error vectors. Each produces $n+1$ distinct insertion outputs before global cross-shell deduplication, yielding

$$Q_{\text{gen}}, Q_{\text{mem}} \le (n+1)\,\mathrm{Vol}(m,S), \qquad Q_{\text{score}} \le Q_{\text{mem}}.$$

The binomial-quantile and expected bounds follow by monotonicity in T and averaging over $T \sim \mathrm{Binomial}(m, p_s)$.

The finite-confidence table in M was independently recomputed and is exact:

n	δ	L_n	τ_δ	$(n+1)\,\mathrm{Vol}(n-1, \tau_\delta + L_n - 1)$
16	10^{-2}	1	3	9,792
16	10^{-3}	1	4	32,997
24	10^{-2}	2	4	1,113,800
24	10^{-3}	2	5	3,637,475
32	10^{-2}	2	5	31,107,417
32	10^{-3}	2	6	117,883,392

3.5 High-probability exponent

Let

$$\epsilon_n = \sqrt{\frac{c \ln n}{2m}}.$$

Hoeffding's inequality gives

$$\Pr\{T > m(p_s + \epsilon_n)\} \le e^{-2m\epsilon_n^2} = n^{-c}.$$

On the complementary event, define the deterministic radius fraction

$$a_n = p_s + \epsilon_n + \frac{L_n - 1}{m} = p_s + o(1).$$

For all sufficiently large n, $a_n < 1/2$ and $S/m \le a_n$. Monotonicity of Hamming-ball volume and of h_2 on $[0, 1/2]$ gives

$$\mathrm{Vol}(m, S) \le 2^{mh_2(a_n)}.$$

Consequently,

$$\frac{1}{n}\log_2 Q_{\mathrm{mem}} \le \frac{\log_2(n+1)}{n} + \frac{m}{n}h_2(a_n) = h_2(p_s) + o(1),$$

and the same bound holds for Q_{score}. If $|\mathcal{C}_n| = 2^{nR+o(n)}$, exhaustive codeword-wise scoring evaluates every codeword, and on the same event

$$\frac{|\mathcal{C}_n|}{Q_{\mathrm{score}}} \ge 2^{n(R-h_2(p_s)-o(1))}.$$

This proves the stated theorem after the written repair identified below.

4 Checklist-by-checklist verdict

1. **PASS**. **Channel law with substitutions before deletion.** S lines 32–60 are correct. Conditional on $J = j$, only the m surviving substitution variables are constrained. The deleted-coordinate substitution is summed out to one. No extra factor of p_s or $1 - p_s$ is missing.

2. **PASS**. **Sliding mismatch recurrence.** M lines 77–88 and S lines 39–59 use a consistent one-based convention. The recurrence removes (x_{j+1}, y_j) from the suffix and adds (x_j, y_j) to the prefix.

3. **PASS**. **Strict reversal, multiplicities, $n = 4$, and equality.** S lines 67–94 are correct. The x_A multiplicities at distances $0, 1, 2, 3$ are $1, 1, 1, n - 3$; at $n = 4$ this is one of each. The factorization is exact. At $np_s = 1 + 2p_s^2$ the aggregate scores tie; above it x_B is strictly aggregate-ML, while x_A remains strictly best-component for every $0 < p_s < 1/2$.

4. **PASS**. **The $n + 1$ one-insertion sphere and classical labeling.** S lines 96–110 are correct, and S lines 107–109 expressly label the result classical and not a new theorem. M lines 112–115 do the same in compressed form.

5. **PASS**. **Both directions of shell characterization.** S lines 111–121 are complete. An alignment of distance s gives an error vector of weight s and reconstructs x by reinsertion; every generated candidate conversely carries such an alignment and vector. Taking the minimum proves first appearance and end-of-shell discovery.

6. **PASS**. **Every alignment of an unseen word has distance at least $s + 1$.** If any $d_j \le s$, then $h_y(x) \le s$ and Proposition 3 would already have generated the word. Integrality gives $d_j \ge s + 1$ for every j.

7. **PASS**. **Strict incumbent comparison is necessary for complete tie-set certification.** S lines 128–139 are correct. The explicit $n = 4$, $p_s = 1/3$, $y = 001$ witness above shows that equality with U_s can occur simultaneously for a seen and an unseen codeword. Non-strict comparison certifies an ML maximizer but not a complete ML tie set.

8. **PASS**. **$p_s = 0$ and $p_s = 1/2$ boundaries.** At $p_s = 0$, a candidate has positive likelihood exactly when at least one alignment has distance zero; every such candidate is in shell zero. If no positive incumbent exists for an impossible observation, exhaustion still returns the all-zero-likelihood tie set. At $p_s = 1/2$, every candidate has likelihood 2^{-m}, so no strict shell decay is available and complete-tie certification requires exhaustion. It would improve presentation to state explicitly the standard Bernoulli-mass convention for the 0^0 terms.

9. **PASS. Definition and floor formula for $L_n(p_s)$.** S lines 144–150 are exact. Since $\rho^\ell < 1/n$ is equivalent to $((1-p_s)/p_s)^\ell > n$, the smallest integer is $\lfloor \log_{(1-p_s)/p_s} n \rfloor + 1$. The formula correctly handles exact logarithmic boundaries because the inequality is strict.

10. **PRECISE REPAIR. Stopping ratio and indexing.** The ratio and the index $s = T + L_n - 1$ are correct *when that shell is below m*. However, S lines 161–165 and M lines 195–198 invoke U_s without first separating the case $T + L_n - 1 \geq m$, even though U_s was defined only for $s < m$.

    Exact repair: let $s_0 = T + L_n - 1$. First state: if $s_0 \geq m$, exhaustion gives $S \leq m = \min\{m, s_0\}$. Otherwise $s_0 < m$, U_{s_0} is defined, and the displayed ratio equals $n\rho^{L_n} < 1$. This is a proof-domain edit only; the theorem and indexing are correct.

11. **PASS. Deterministic, finite-confidence, and expected query bounds.** S lines 168–191 follow from counting all shells through S, global deduplication, and $Q_{\text{score}} \leq Q_{\text{mem}}$. The finite-confidence bound follows on $T \leq \tau_\delta$, and the expected bound is the exact binomial average of the deterministic cap. The six numerical caps in M Table I were independently recomputed and match.

12. **PRECISE REPAIR. Hoeffding/Hamming-ball argument and the $o(1)$ transition.** The theorem is correct, but S line 207 writes

    $$2^{mh_2(S/m)} = 2^{n(h_2(p_s)+o(1))}.$$

    Only the upper bound $S/m \leq p_s + o(1)$ has been proved; S/m need not converge to p_s, so that equality is not justified.

    Exact repair: introduce $a_n = p_s + \epsilon_n + (L_n - 1)/m$. On the Hoeffding event, $S/m \leq a_n$, $a_n = p_s + o(1)$, and eventually $a_n < 1/2$. Then write

    $$\mathrm{Vol}(m, S) \leq 2^{mh_2(a_n)}$$

    and use continuity of h_2, $m/n \to 1$, and $\log_2(n+1)/n \to 0$. M lines 209–213 should add this monotonicity-and-continuity sentence. This repair changes proof presentation only, not the exponent.

13. **PASS. Rate comparison baseline.** S lines 195–210 and M lines 201–214 compare complete candidate-likelihood evaluations against exhaustive *codeword-wise* scoring. They do not claim dominance over all exact decoders or over code-specific algorithms, and they expressly retain the membership-oracle scope.

14. **PASS. No hidden linearity, parity-check, or unique-ML assumption.** Every proof uses only nonempty code membership and the fact that the transmitted word lies in the code. No linearity, syndrome cost, parity-check matrix, or codeword enumeration is used in the correctness proof. Strict stopping returns the complete ML tie set; uniqueness is not assumed.

15. **PASS. Empirical representative disagreements are not presented as strict pathwise failure.** M lines 305–311 explicitly state that the sampled tie sets always overlapped and that representative differences are not evidence of strict pathwise suboptimality. The strict claim rests solely on Theorem 1.

5 Additional presentation clarification

S lines 63–65 define $M_y(x)$ as the maximum *alignment component* after the substitution at the deleted coordinate has been marginalized. A fully specified pre-deletion substitution history contains one additional latent bit. For $0 < p_s < 1/2$, maximizing that bit multiplies every candidate's alignment score by the same factor $1 - p_s$, so all orderings and the reversal theorem are unchanged.

To remove any semantic ambiguity, replace "most likely individual compatible history" by "largest deletion-alignment component" or add one sentence explaining the common deleted-coordinate factor. This is presentation only.

6 Defect classification and final decision

Issue	**Effect**	**Classification**
Stopping-proof case split	U_s must not be invoked outside its declared range $s < m$. The theorem remains valid by exhaustion.	Presentation/proof-domain only
High-probability equality	Replace an unjustified equality by a monotone upper bound at deterministic radius a_n. The exponent is unchanged.	Presentation/proof-rigor only
"History" terminology	M_y is an alignment-component maximum; full-history ordering differs only by a common factor.	Presentation only
Proposal score-before-membership order	Unnecessarily obscures the achieved separation between membership calls and complete scoring.	Proposal integration/efficiency only

No identified issue changes mathematical correctness or stated scope. After the two required proof edits, every checklist obligation passes.

$\boxed{\textbf{PASS\_WITH\_EDITS}}$

Final mathematical recommendation: apply the two exact proof repairs in both the conference manuscript and proof supplement, make the optional terminology clarification, and retain all theorem statements and numerical claims within their present scope. No new simulation is required to resolve these mathematical edits.